

Week2_1st_Professional Software Engineering_Lecture

Professional Software Engineering 课堂笔记与翻译

Professional Software Engineering - Week 1 Session

Introduction & Basic Concepts

Mohammad: This is for your heat up, this is all of the things that we need during the next 14 weeks.

Mohammad: I ask you for the basic concept we need to get ourselves coming here.

1. INDENTATION

Mohammad: The first one, the first basic concept is an **indentation**.

Mohammad: Like what? It's a kind of the tab.

Mohammad: If you're gonna write the program, for example, in C++,

- usually you define the boundary of each function or method.
- The starting syntax and the last syntax that belongs to that method.

Mohammad: they call it the body of the function is that correct?

Mohammad: In Python, usually we use **indentation**.

 **Mohammad:** Anyone doesn't know about indentation?

Mohammad: For example, we're gonna write a simple function to do the recursive method using the POW.

Mohammad: what was the pw exam one it's power one you can write it as a requested function.

Mohammad: how many syntax will be the body of that function maybe three four it depends on the title your body.

Mohammad: how the compiler the python compiler can recognize how many syntax it belongs to that method based on the engine we're gonna have a look.

Mohammad: if you miss this slide is available in week one you can download this slide I think I mentioned it.

Mohammad: you can see this space I'm alright as a tab this will help the compiler to understand how many syntax are in the body of the function.

Mohammad: the functions maybe for example simple print type and just there is one.

Mohammad: if you have many more syntax all should follow exactly the same if you go back walk one happy it will not be lost that function.

 **Mohammad:** any question for how good let's go back to the outline for the first week.

2. VARIABLE NAMING

Mohammad: different type of variable if we want to put a name we have the flexibility the same as other voting language.

 **Mohammad:** can we start the name of the variable with a number?

Mohammad: no we have some rule that we need to follow when you want to choose the name of the variable.

Mohammad: of course it should be meaningful don't use x and y z w for the name of the variable for the coding as a software developer.

3. CODE READABILITY & DOCUMENTATION

Mohammad: whenever we want to develop the code your coding should be readable for someone else.

Mohammad: of course you're gonna add comments for your for each module.

Mohammad: later on we will discuss we can write it down manually or some of some parts have been done.

Mohammad: but generally speaking for all of the projects that you're going to work you need to get used to that.

Mohammad: because i saw this issue in the past years of experience developers are not willing to write or you then get used to that to write a comment.

Mohammad: because the documentation is the key for the future development.

Mohammad: if you leave it behind it will be really tough for someone else as a team work gonna contribute on that.

Mohammad: it means we don't want to tie the project to a person as a developer because we want to work as a team.

Mohammad: if you write it down the code is meaningful it's easy to read and the name of the variable functions or methods all classes whatever that you're gonna use it is relevant to the project.

Mohammad: plus comment will help you as a software developer for future updates.

Mohammad: it doesn't mean someone else gonna take the forward take over of your work of course people continue your work easily if you have the clear and clean documentation.

 **Mohammad:** Is it your proficiency? If

专业软件工程 - 第 1 周课程

介绍与基本概念

Mohammad: 这是为你们的热身，这些是我们在接下来 14 周内需要的所有内容。

Mohammad: 我要求你们掌握我们需要了解的基本概念。

1. 缩进

Mohammad: 第一个，第一个基本概念是**缩进**。

Mohammad: 像什么？它是一种制表符。

Mohammad: 如果你要写程序，例如用 C++，

- 通常你会定义每个函数或方法的边界。
- 属于该方法的起始语法和最后语法。

Mohammad: 他们称之为函数体，对吗？

Mohammad: 在 Python 中，我们通常使用**缩进**。

 **Mohammad:** 有人不知道缩进吗？

Mohammad: 例如，我们将编写一个简单的函数，使用 POW 来实现递归方法。

Mohammad: 那个 pw 示例是什么，是 power 函数，你可以把它写成一个请求的函数。

Mohammad: 那个函数体将有多少行语法，可能三行四行，这取决于你的函数体内容。

Mohammad: Python 编译器如何识别有多少行语法属于那个方法，基于引擎，我们稍后会看一下。

Mohammad: 如果你错过了这张幻灯片，它在第一周的资料里，你可以下载这张幻灯片，我想我提到过。

Mohammad: 你可以看到这个空格，我把它当作一个制表符，这将帮助编译器理解函数体中有多少行语法。

Mohammad: 函数可能例如只是一个简单的打印类型，只有一行。

Mohammad: 如果你有更多行语法，所有行都应遵循完全相同的缩进，如果你回退一个缩进层级，它就不会属于那个函数了。

 **Mohammad:** 关于这个有什么问题吗？让我们回到第一周的课程大纲。

2. 变量命名

Mohammad: 不同类型的变量，如果我们想给它起个名字，我们和其他编程语言一样有灵活性。

 **Mohammad:** 我们可以用数字作为变量名的开头吗？

Mohammad: 不可以，我们在选择变量名时需要遵循一些规则。

Mohammad: 当然，它应该是有意义的，不要用 x、y、z、w 作为变量名来进行编码，作为一名软件开发人员。

3. 代码可读性与文档

Mohammad: 每当我们开发代码时，你的代码应该对他人来说是可读的。

Mohammad: 当然，你会为你的每个模块添加注释。

Mohammad: 稍后我们会讨论，我们可以手动写下来，或者有些部分已经完成了。

Mohammad: 但总的来说，对于你将参与的所有项目，你需要习惯这一点。

Mohammad: 因为根据我过去几年的经验，我看到这个问题，开发人员不愿意写注释，或者你还没有习惯写注释。

Mohammad: 因为文档是未来开发的关键。

Mohammad: 如果你把它抛在脑后，对于团队中其他将要参与工作的人来说，将会非常困难。

Mohammad: 这意味着我们不想把项目绑定在某个开发人员个人身上，因为我们希望以团队形式工作。

Mohammad: 如果你写下来，代码是有意义的，易于阅读，并且变量、函数或方法、所有类等，无论你使用什么，其名称都与项目相关。

Mohammad: 此外，注释将帮助你作为一名软件开发人员进行未来的更新。

Mohammad: 这并不意味着别人会直接接管你的工作，当然，如果你有清晰干净的文档，别人可以轻松地继续你的工作。

 **Mohammad:** 这是你的专长吗？如果

Mohammad: If you want to work in a professional way, we need to do it.

We have a couple of different tools that can help you in industry to write the proper documentation for your project.

But one of the mandatory parts is **inline comments**. Later on, we will see inline comments.

It means in your coding step, you need to explain:

- What is the rule?
- Or what is the necessity of this method?
- Why you are writing this class?

Maybe in one or two lines, you can explain. If someone else just has a look at your code, they don't need to go into the detail of your project to understand what this method is for and its usage.

This is because of development, extension, and many more things.

Okay, the input and output I think is clear for your project. Of course, all of the project should have output.

Any module that you're going to run, some of them have end-user input (you need to enter something), some others don't need it. It depends on the module that you're going to run.

We talked about the comments. We will talk more about different data types.

Data Types like integer, float, double, character, boolean, and then later on we're going to see more complex data types like 2D, 3D, tensors, arrays.

Last but not least, **mathematical operations**.

Please review all of the slides if you have any questions.

 **Student:** Do I need to explain this as an example?  **Mohammad:** Okay, what is the type of this? What is the complex type?

 **Student:** Okay, how about integer? Give me an example to make sure we are on the same page.  **Mohammad:** We are talking about the data type `int` for integer. For any other languages that you worked with before, maybe in the past, like C programming,

C++, C#, Java, MATLAB, R programming, Python, whatever, we have different types. Right in between integer, you can see the example.

👤 **Student:** Okay, float? 🤖 **Mohammad:** Yeah, we have this. It's a number, positive or negative, it doesn't matter.

And we have the complex. It is not limited to that, but have a look. Complex means, for example, `1 + 2j`.

If you want to understand the mathematics behind that, you need to do a little bit of research to understand what is the concept behind the complex variable.

But it is the type of the variable. If you don't know, please do a little bit of research to understand what it is, because I cannot explain what the math concept is. It's a simple thing for the complex.

You need to do the research to understand the mathematical concept behind that. If you want to calculate the value of the complex value, how we can calculate it?

A little bit more about the complex value, we will see in the next couple of weeks. We just want to use it in a different project.

But keeping in mind, at the Master's level, always your **research capability** is your winning card.

It means, what's the difference between undergraduate and the Master's? You can efficiently use the research.

Mohammad: 如果你想以专业的方式工作，我们就需要这样做。

我们有一些不同的工具可以帮助你为工业界的项目编写适当的文档。

但其中一个强制性的部分是**行内注释**。稍后，我们会看到行内注释。

这意味着在你的编码步骤中，你需要解释：

- 规则是什么？
- 或者这个方法的必要性是什么？
- 你为什么要写这个类？

也许用一两行，你就可以解释清楚。如果别人只是看一眼你的代码，他们不需要深入了解你的项目就能明白这个方法是做什么的以及它的用途。

这是因为开发、扩展以及许多其他原因。

好的，我认为输入和输出对你的项目来说是清楚的。当然，所有项目都应该有输出。

任何你要运行的模块，其中一些有最终用户输入（你需要输入一些东西），另一些则不需要。这取决于你要运行的模块。

我们讨论了注释。我们将更多地讨论不同的数据类型。

数据类型，如整数、浮点数、双精度浮点数、字符、布尔值，然后稍后我们会看到更复杂的数据类型，如二维、三维、张量、数组。

最后但同样重要的是，**数学运算**。

如果你有任何问题，请复习所有的幻灯片。

🧑‍🎓 **学生**：我需要把这个作为一个例子来解释吗？🧑‍🎓 **Mohammad**：好的，这个的类型是什么？什么是复数类型？

🧑‍🎓 **学生**：好的，那整数呢？给我一个例子，以确保我们在同一页上。🧑‍🎓 **Mohammad**：我们讨论的是整数数据类型 `int`。对于你以前用过的任何其他语言，也许过去用过，比如 C 语言、C++、C#、Java、MATLAB、R 语言、Python 等等，我们都有不同的类型。在整数之间，你可以看到例子。

🧑‍🎓 **学生**：好的，浮点数？🧑‍🎓 **Mohammad**：是的，我们有这个。它是一个数字，正数或负数，这没关系。

我们还有复数。它不限于此，但请看。复数意味着，例如，`1 + 2j`。

如果你想理解这背后的数学原理，你需要做一点研究来理解复数变量背后的概念是什么。

但它是变量的类型。如果你不知道，请做一点研究来理解它是什么，因为我无法解释数学概念是什么。对于复数来说，这是一个简单的事情。

你需要做研究来理解其背后的数学概念。如果你想计算复数的值，我们该如何计算？

关于复数值的更多内容，我们将在接下来的几周内看到。我们只是想在另一个项目中使用它。

但要记住，在硕士阶段，你的**研究能力**始终是你的制胜法宝。

这意味着，本科和硕士的区别是什么？你可以有效地利用研究。

1. ABSOLUTE FIDELITY (ENGLISH)

To calculate the value of the complex value, how we can... A little bit more about the complex value we will see in the next couple of weeks. We just want to use it in a different project.

But keeping your mind at the master level: always, your **research capability** is your winning card.

It means, what's the difference between undergrad and the master? You can efficiently use the research.

And what's the difference between the PhD and master? PhD can work on the project no one else in the world has worked on before.

At the master level, we don't expect you to solve a problem that no one has ever worked on before. But still, you have the capability to use the current solution efficiently, and find the right solution—an efficient one—in the limited time.

Because undergrads, usually, they will do their research. Of course, they can find it through different Google engines, AI tools, and many more. But at the master level, you should save the time to find a solution for your question, whatever it is, from the simple to the really complex one.

All of the resources are available online publicly as open access, open source. But who can use it efficiently is the difference—the main difference, or one of the most important differences—between the undergrad and the master project or expectation.

That's why it's a simple concept behind the complex variable. You need to do a little bit of research to understand what it means as a concept behind that.

It's like the factorial. Do you know what it means? Factorial, for example, to calculate for n as an integer. Just similar things. It's a mathematical concept. You understood it, then you can code it.

We have the simple complex concept. How about the boolean? How many more? Please take a look at that. Definitely, we need these sequence types, of course, meaning more in the next projects: as a string, as a list... A couple we will not leave these things behind because on a daily basis we're gonna use it.

We need to know what is the data structure for each variable that we're gonna use, and what is a suitable data type, not as a random error to use.

I'm mapping type, they call it dictionary. In the dictionary, we have the value and key. It's somehow a simulation of the data tables in the databases.

For example, you have a couple of tables in the database. In each table, we have a couple of features. It means columns. Each column has a name. Here we have the key value and also a name.

We already started again for the input, output, and processing. We don't need to go more in detail.

 **Student:** Can you give an example for any project that we don't need input value?

 **Mohammad:** For example, we want to have the method to return the current time. Do we need input? Automatically, we can just generate what is the current time based on the zone, the different region.

 **Student:** Can we have any method without output? Does it make sense?

 **Mohammad:** We have to because you're developing the different module function... Well, get something out of that. That's right. If you're gonna write something and there is no outcome in a...

5. 中文全文翻译

要计算复数的值，我们该如何...关于复数值的更多内容，我们将在接下来的几周内看到。我们只是想在另一个项目中使用它。

但要记住，在硕士阶段：始终，你的**研究能力**是你的制胜法宝。

这意味着，本科和硕士的区别是什么？你可以有效地利用研究。

那么博士和硕士的区别是什么？博士可以研究世界上此前无人涉足过的项目。

在硕士阶段，我们不期望你去解决一个从未有人研究过的问题。但你仍然有能力高效地利用现有的解决方案，并在有限的时间内找到正确的、高效的解决方案。

因为本科生通常也会进行研究。当然，他们可以通过不同的谷歌搜索引擎、AI工具等找到信息。但在硕士阶段，你应该节省时间为你的问题找到解决方案，无论问题是简单还是极其复杂。

所有的资源都可以在线公开获取，作为开放获取、开源资源。但谁能高效地利用它，这就是区别——主要的区别，或者说本科和硕士在项目或期望上最重要的区别之一。

这就是为什么复数变量背后是一个简单的概念。你需要做一点研究来理解其背后的概念是什么。

这就像阶乘。你知道它是什么意思吗？例如，计算整数 `n` 的阶乘。类似的事情。这是一个数学概念。你理解了它，然后你就能编写代码。

我们有简单的复数概念。布尔型呢？还有多少种？请看一看。当然，我们肯定需要这些序列类型，在接下来的项目中意义更大：比如字符串、列表... 有几种我们不会遗漏，因为日常工作中我们会用到它们。

我们需要知道我们将要使用的每个变量的数据结构是什么，以及什么是合适的数据类型，而不是随意错误地使用。

映射类型，他们称之为字典。在字典中，我们有值和键。它在某种程度上模拟了数据库中的数据表。

例如，你在数据库中有几个表。在每个表中，我们有几个特征。也就是列。每列都有一个名称。这里我们有键值，也有名称。

我们已经再次讨论了输入、输出和处理。我们不需要更详细地讨论。

 **学生：**你能举一个我们不需要输入值的项目的例子吗？

 **Mohammad：**例如，我们想要一个返回当前时间的方法。我们需要输入吗？自动地，我们可以根据时区、不同地区生成当前时间。

 **学生：**我们可以有任何没有输出的方法吗？这有意义吗？

 **Mohammad：**我们必须有，因为你正在开发不同的模块功能... 嗯，从中得到一些东西。没错。如果你要写一些东西，而其中没有结果...

1. ABSOLUTE FIDELITY (ENGLISH)

Mohammad: ...automatically we can just generate what is the current time based on the zone, the different region.

 **Student:** Can we have any method without output? Does it make sense?

 **Mohammad:** We have to because you're developing the different module function... well, get something out of that. That's right. If you're gonna write something and there is no outcome in all circumstances, then there is no usage for that.

Keep in your mind always: whatever that you're gonna write as a method or function, even a small one, we should expect in some cases at least to have an outcome. Without outcome in all conditions, there is no matter to write. But we can't escape for the image for some of the cases.

Okay, I expect you to review all of these things for the determination for the printing. The simple: if you want to print out the outcome in the new line, what you need to run?

 **Student:** What is the function?

 **Mohammad:** Okay, is that function automatically available? Just you can use it and your project is waiting to get an input.

We're gonna start our session with one simple project. I want to make sure you understand all of these concepts: which one as an input, output, different type of the variable, and many more. If you miss someone, at least you can learn. Okay? To make sure all of us we are in the same page. If we don't have the background, definitely we need to read.

 **Student:** Where is their source that we need to read? Do you have access to the PDF file?

 **Mohammad:** No, the Python coding... yes, no, not this one. As an ebook, we have access to the Python table? Where is that? Did you check it initially? I can hear you. Okay, see this? I'm going to make it a little bit larger. Is it things that you can see in your blackboard? Am I right? Your elements. Or is it something else?

Today we're gonna talk about your assessment, especially the first assessment. Briefly, to have a better understanding what we are approaching.

We are not learning the concept just to do the assessment. But we are trying to learn as many as possible different tools, concepts behind the object-oriented programming to do the project for the assessment and activity in the best way. Okay?

You will learn more than that, but up to you how you gonna manage your assessment submission or activity submission. How much you want to go beyond that the minimum requirement.

First week, we can see this one. Is that correct? Is the PowerPoint that we are talking with this first about set up your environment.

Anyone has been successfully installed on a panda or VS code? It all good. Please don't finish up. If you face any difficulty, ask me. We will go with each other too.

So now, we try to learn one function or one tool. Is that freeze? Really, it will help you to generate the requirements that takes part for any project that you're going to develop from now on.

When you need to run this at the end, when you want to deploy it, means the last step. Because during the development, you are adding more packages, libraries to your project. If you run it in the...

2. STRUCTURAL REORGANIZATION (ENGLISH)

Mohammad: ...automatically we can just generate what is the current time based on the zone, the different region.

 **Student:** Can we have any method without output? Does it make sense?

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We're gonna start our session with one simple project. I want to make sure you understand all of these concepts:

- Which one as an input, output.
- Different type of the variable.
- And many more.

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Transition to Assessment Discussion

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First Week & Environment Setup

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Anyone has been successfully installed on a panda or VS code? It all good. Please don't finish up. If you face any difficulty, ask me. We will go with each other too.

Introduction to a Tool: **freeze**

So now, we try to learn one function or one tool. Is that **freeze** ? Really, it will help you to generate the **requirements** that takes part for any project that you're going to develop from now on.

When you need to run this at the end, when you want to deploy it, means the last step. Because during the development, you are adding more packages, libraries to your project. If you run it in the...

3. LOGICAL HIGHLIGHTING (ENGLISH)

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If you're gonna write something and there is no outcome in all circumstances, then there is no usage for that.

Key Principle: Keep in your mind always: whatever that you're gonna write as a method or function, even a small one, we should expect in some cases at least to have an outcome.

Without outcome in all conditions, there is no matter to write. But we can't escape for the image for some of the cases.

Okay, I expect you to review all of these things for the determination for the printing.

Example Prompt: The simple: if you want to print out the outcome in the new line, what you need to run?

 **Student:** What is the function?

 **Mohammad:** Okay, is that function automatically available? Just you can use it and your project is waiting to get an input.

Session Objective: Simple Project

We're gonna start our session with one simple project. I want to make sure you understand all of these concepts:

- Which one as an **input, output**.
- Different **type of the variable**.
- And many more.

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PART 2: COURSE ASSESSMENT & TOOLS

2.1 Discussion on the First Assessment

Today we're gonna talk about your **assessment**, especially the **first assessment**. Briefly, to have a better understanding what we are approaching.

We are not learning the concept just to do the assessment. But we are trying to learn as many as possible different tools, concepts behind the **object-oriented programming** to do the project for the assessment and activity in the *best way*. Okay?

You will learn more than that, but up to you how you gonna manage your assessment submission or activity submission. How much you want to go beyond that the *minimum requirement*.

2.2 First Week Recap & Environment Check

First week, we can see this one. Is that correct? Is the PowerPoint that we are talking with this first about **set up your environment**.

Anyone has been successfully installed on a **panda** or **VS code**? It all good. Please don't finish up. If you face any difficulty, ask me. We will go with each other too.

2.3 Introducing a Development Tool: **pip freeze**

So now, we try to learn one function or one tool. Is that **freeze** ? Really, it will help you to generate the **requirements** that takes part for any project that you're going to develop from now on.

When you need to run this at the end, when you want to deploy it, means the last step. Because during the development, you are adding more **packages, libraries** to your project. If you run it in the...

5. FULL CHINESE TRANSLATION

Mohammad:我们可以根据时区、不同地区自动生成当前时间。

 **学生:** 我们可以有任何没有输出的方法吗? 这有意义吗?

 **Mohammad:** 我们必须有, 因为你正在开发不同的模块功能.....嗯, 从中得到一些东西。没错。

如果你要写一些东西, 而在所有情况下都没有结果, 那么它就没有任何用处。

核心原则: 请始终牢记: 无论你将编写什么方法或函数, 即使是一个很小的, 我们都应该预期至少在某些情况下会有结果。

如果在所有条件下都没有结果, 那就没有编写的必要。但对于某些情况下的形象, 我们无法回避。

好的, 我希望你们为了确定打印内容, 复习所有这些内容。

示例提示: 简单来说: 如果你想在新的一行打印出结果, 你需要运行什么?

 **学生:** 函数是什么?

 **Mohammad:** 好的, 那个函数是自动可用的吗? 你只需使用它, 你的项目正在等待获取输入。

本节目标: 简单项目

我们将从一个简单的项目开始本次课程。我想确保你们理解所有这些概念：

- 哪些作为**输入、输出**。
- 不同的**变量类型**。
- 以及更多。

如果有人错过了，至少你可以学习。好吗？为了确保我们所有人都在同一页面上。如果我们没有相关背景，我们肯定需要阅读。

 **学生**：我们需要阅读的资料来源在哪里？你们能访问PDF文件吗？

 **Mohammad**：不，Python编码.....是的，不，不是这个。作为电子书，我们可以访问Python表格？那个在哪里？你最初检查过吗？

 **学生**：我能听到你。

 **Mohammad**：好的，看到这个了吗？我要把它放大一点。这是你在黑板上能看到的東西吗？我说得对吗？你的元素。还是别的什么？

第二部分：课程评估与工具

2.1 关于第一次评估的讨论

今天我们将讨论你们的**评估**，特别是**第一次评估**。简要地说，以便更好地理解我们将要做什么。

我们学习概念不仅仅是为了完成评估。但我们正试图尽可能多地学习不同的工具、**面向对象编程**背后的概念，以便以**最佳方式**完成评估项目和活动。好吗？

你们将学到更多，但如何管理你们的评估提交或活动提交取决于你们自己。你们想在**最低要求**之外走多远。

2.2 第一周回顾与环境检查

第一周，我们可以看到这个。对吗？我们首先讨论的是关于**设置你的环境**的PowerPoint。

有没有人成功安装了**panda**或**VS code**？都很好。请不要就此结束。如果遇到任何困难，请问我。我们也会互相帮助。

2.3 介绍一个开发工具: `pip freeze`

那么现在，我们尝试学习一个函数或一个工具。是 `freeze` 吗？真的，它将帮助你生成从现在开始你将开发的任何项目所需的**依赖项 (requirements)**。

当你需要在最后运行这个，当你想部署它时，意味着最后一步。因为在开发过程中，你正在向你的项目添加更多的**包、库**。如果你在.....中运行它。

2.3 INTRODUCING A DEVELOPMENT TOOL: `PIP FREEZE`

Now, we will try to learn a function or a tool. It is `freeze`. It will help you generate the **requirements** for any project you are going to develop from now on.

You need to run this at the end, when you want to deploy it. This is the last step because during development, you are adding more **packages** and **libraries** to your project.

If you run it in the middle, some of them will be missing, and the final part is the crucial one.

2.4 COURSE MATERIAL GUIDANCE

The material I am asking you to cover goes from simple to advanced.

- **Chapter 2** is a good start for those who are a bit rusty with coding.
- If you are not familiar, start with **Chapter 1**. It sets up the basics.
- If you are familiar with other languages like Java, then you should start from **Chapter 2**. It's simple, but just review it quickly.
- You don't need to spend too much time on Chapter 1; it's up to you. I will leave it to you.

For **Chapter 1** and **Chapter 3** as well, you need to gradually keep them as your reference, not just for week one. We will need them for the next couple of weeks.

I will share more resources. You are going to have access soon, today.

 **Student:** I'm not sure if we have access. Maybe you have access? Let's double-check with each other.  **Mohammad:** I forgot. You are going to have access to these. It should be the same. Let's look at your Elements. Do you have access? You don't? Okay, let me refresh from my side.

 **Student:** I don't have access to Reach One, but I have access to... sorry, which...  **Mohammad:** Okay, and then you need to fill up that form. Did you do it? How about you?

Please do it if you don't have access to any parts of the resources we are talking about.

For example, for **Elements**, you need to fill up that form that was shared, I think, yesterday or a couple of days before. Please fill it up. Otherwise, it will be a little bit time-consuming to get access.

Just two of you don't have access? The rest are okay? I see.

Because a couple of you still don't have access, I will share the resources today on Teams as well to make life easier. But hopefully, from next week, it will be sorted out, and all of us will save time to go forward.

I shared these ebooks and the themes last week. Okay, all of the resources that we wanted, I have shared for those who don't have access.

2.5 WEEK 1 & 2 RECAP AND ASSESSMENT OVERVIEW

Okay, these are the things that you should see so far for week 1 and 2 concepts, before going for Activity One, to make sure you are aware of the coding.

Let's talk about the deadline for week one. Now we are in week two. Is that correct? Week two.

The deadline for **Assessment One** is **week seven**.

If you have a look at the left side, there is a section for **Assessment Submission**. You can see the weight of each assessment.

If you scroll down, it's for the different courses: MSA 800 for your research methods, or your fossil computing for different courses.

- How many assessments for MSA 800? For Professional Software Development, it is **three**.
- The rest of the courses, except the final test, have **two**.
- Just MSA 800 and also the final test, you have three official submissions.

2.3 介绍一个开发工具： PIP FREEZE

现在，我们将尝试学习一个函数或一个工具。它就是 `freeze`。它将帮助你生成从现在开始你将开发的任何项目所需的**依赖项 (requirements)**。

你需要在最后运行这个命令，当你想部署它的时候。这是最后一步，因为在开发过程中，你正在向你的项目添加更多的**包和库**。

如果你在中间运行它，其中一些将会缺失，而最后的部分才是关键。

2.4 课程材料指导

我要求你们学习的材料是从简单到高级的。

- **第 2 章** 对于编码有些生疏的人来说是一个好的起点。
- 如果你不熟悉，从**第 1 章**开始。它建立了基础知识。
- 如果你熟悉其他语言，如 Java，那么你应该从**第 2 章**开始。它很简单，但请快速复习一下。
- 你不需要在第一章上花太多时间；这取决于你。我把它留给你决定。

对于**第 1 章**和**第 3 章**也是如此，你需要逐步将它们作为你的参考资料，而不仅仅是为了第一周。我们在接下来的几周里也会需要它们。

我会分享更多资源。你们很快就会获得访问权限，今天。

 **学生**：我不确定我们是否有访问权限。也许你有权限？我们互相确认一下。 

Mohammad：我忘了。你们将会获得这些资源的访问权限。应该是一样的。我们看看你的 Elements。你有访问权限吗？你没有？好的，让我从我这端刷新一下。

 **学生**：我无法访问 Reach One，但我可以访问.....抱歉，哪个.....  **Mohammad**：好的，然后你需要填写那个表格。你填了吗？你呢？如果你无法访问我们正在讨论的任何资源部分，请填写它。

例如，对于 **Elements**，你需要填写那个分享的表格，我想是昨天或几天前分享的。请填写它。否则，获得访问权限会有点耗时。

只有你们两个没有访问权限吗？其他人都有吗？我明白了。

因为你们中还有几个人没有访问权限，我今天也会在 Teams 上分享资源，以便让事情更容易些。但希望从下周开始，这个问题会得到解决，我们所有人都能节省时间继续前进。

我上周分享了这些电子书和主题。好的，所有我们想要的资源，我已经为那些没有访问权限的人分享了。

2.5 第 1 周和第 2 周回顾与评估概述

好的，这些是到目前为止，在进行活动一之前，你们应该看到的第 1 周和第 2 周的概念内容，以确保你们了解编码。

我们来谈谈第一周的截止日期。现在我们在第二周。对吗？第二周。

评估一的截止日期是**第七周**。

如果你看一下左侧，有一个**评估提交**部分。你可以看到每个评估的权重。

如果你向下滚动，它是针对不同课程的：MSA 800 用于你的研究方法，或者其他课程的高性能计算。

- MSA 800 有多少个评估？对于专业软件开发，是**三个**。
- 其余课程，除了期末考试，有**两个**。
- 只有 MSA 800 以及期末考试，你有三个正式提交。

1. ABSOLUTE FIDELITY (ENGLISH ONLY)

If you scroll down, it's for the different courses: MSA 800 for your research methods, or high-performance computing for different courses.

How many assessments for MSA 800? For professional software development, it is **three**.

The rest of the courses, except the final test, have **two**.

Only MSA 800 and also the final test, you have three official submissions.

Don't stop because sometimes you will keep in your mind, for example for quantum we have two submissions, then you will miss one of the deadlines.

For MSA 800, the first assessment is related to the current system where you can get it. I'm gonna show it to you.

On the top left side, there is a **Programming Class Assessment Info** in your platform.

In this page, you have the assessment info. Please click on that.

Then you can see MSA 800. There is a general detailed explanation.

And then more detail for the first assessment. There is a link. You need to click on that.

As a PDF file, you can see this one. If you click on that, you will see the detail of the requirement for your assessment one, and the scenario behind the assessment, the submission minimum requirement, the things that you need to submit by deadline, all of the requirements for different parts.

The task that we need to consider is to take time to read it carefully.

And then finally, there will be, for example, the requirement if you want to make sure if you want to just be in the minimum requirement task, then that I don't recommend is a C. C means 50 plus. You cannot be less than 50.

Alright, let's start with people. Every assessment for all courses, but the requirement for the rubric, being the category A, B, or C, is written here for more clarification.

It will take time to read it. That's why I just open this for you to read it carefully later on in their week. And then if you have any questions, you can ask me.

Is the assessment detail that you need to take time and really possibly record again back to that. We will discuss if you have any difficulty questions we can answer.

Okay, so we try to navigate these access to be easier for you to see is your elements simple submission.

There is a section for the submission when you want to submit. How many times you can submit? Three times maximum. Yes, three times is the maximum number of submissions before the deadline.

For example, by the end of week 7, you have the deadline for the first assessment.

For any reason, you miss one of the files for the submission for the first assessment. You're gonna resubmit this file. The file will be overwritten on top of the previous.

But how many times can you do that? Maximum three times.

It means the first time you made a mistake is okay, you're gonna do it again. And again for the second time you realize that maybe one of the report as a text file or requirement is missing, again you're gonna submit the final submission. Before the deadline, it will not be considered a...

2. STRUCTURAL REORGANIZATION (ENGLISH ONLY)

Assessment Details and Navigation

If you scroll down, it's for the different courses:

- MSA 800 for your research methods.
- High-performance computing for different courses.

Number of Assessments:

- For MSA 800 (Professional Software Development), it is **three**.
- The rest of the courses, except the final test, have **two**.
- Only MSA 800 and the final test have three official submissions.

Don't stop tracking because sometimes you might remember incorrectly. For example, for quantum computing, we have two submissions, and you might miss one of the deadlines.

Accessing Assessment Information (MSA 800 Example)

For MSA 800, the first assessment details are in the system. I will show you.

On the top left side of your platform, there is a section called **Programming Class Assessment Info**.

On that page, you have the assessment information. Please click on that.

Then you can see MSA 800. There is a general detailed explanation.

For more detail on the first assessment, there is a link. You need to click on that link.

You will see a PDF file. If you click on it, you will see:

- The detailed requirements for Assessment One.
- The scenario behind the assessment.
- The submission minimum requirements.
- The items you need to submit by the deadline.
- All requirements for different parts.

The key task is to take time to read it carefully.

Grading Clarification

Finally, there are requirements. If you aim only for the minimum requirement, which I do not recommend, that is a C grade. C means 50 plus. You cannot score less than 50.

Alright, let's proceed. For every assessment in all courses, the rubric requirements for categories A, B, or C are written there for clarification.

It will take time to read. That's why I am opening this for you to read carefully later in the week. If you have any questions, you can ask me then.

This assessment detail requires your time and you may need to revisit it. We will discuss and answer any difficult questions you have.

Submission Process and Limits

Okay, we are trying to make navigating this easier for you to see your submission elements.

There is a submission section. How many times can you submit? A maximum of three times. Yes, three times is the maximum number of submissions allowed before the deadline.

For example, the deadline for the first assessment is the end of Week 7.

If, for any reason, you miss one of the required files for the first assessment submission, you will resubmit that file. The new file will overwrite the previous one.

But how many times can you do this? A maximum of three times.

This means:

1. The first time you make a mistake is okay; you can resubmit.
2. If you realize a second time that maybe a report text file or requirement is missing, you can submit again.
3. You can submit your final submission before the deadline.

3. LOGICAL HIGHLIGHTING (ENGLISH ONLY)

Assessment Details and Navigation

If you scroll down, it's for the different courses:

- **MSA 800** for your research methods.
- High-performance computing for different courses.

Number of Assessments:

- For **MSA 800** (Professional Software Development), it is **three**.
- The rest of the courses, except the final test, have **two**.
- Only **MSA 800** and the final test have three official submissions.

Don't stop tracking because sometimes you might remember incorrectly. For example, for quantum computing, we have two submissions, and you might miss one of the deadlines.

Accessing Assessment Information (MSA 800 Example)

For **MSA 800**, the first assessment details are in the system. I will show you.

On the top left side of your platform, there is a section called **Programming Class Assessment Info**.

On that page, you have the assessment information. Please click on that.

Then you can see **MSA 800**. There is a general detailed explanation.

For more detail on the first assessment, there is a link. You need to click on that link.

You will see a PDF file. If you click on it, you will see:

- The detailed requirements for **Assessment One**.
- The scenario behind the assessment.
- The submission **minimum requirements**.
- The items you need to submit by the deadline.
- All requirements for different parts.

The key task is to *take time to read it carefully*.

Grading Clarification

Finally, there are requirements. If you aim only for the **minimum requirement**, which I do not recommend, that is a **C** grade. **C means 50 plus**. You cannot score less than 50.

Alright, let's proceed. For every assessment in all courses, the rubric requirements for categories **A, B, or C** are written there for clarification.

It will take time to read. That's why I am opening this for you to read carefully later in the week. If you have any questions, you can ask me then.

This assessment detail requires your time and you may need to revisit it. We will discuss and answer any difficult questions you have.

Submission Process and Limits

Okay, we are trying to make navigating this easier for you to see your submission elements.

There is a submission section. **How many times can you submit?** A maximum of **three** times. Yes, **three times is the maximum number of submissions** allowed before the deadline.

For example, the deadline for the first assessment is the end of **Week 7**.

If, for any reason, you miss one of the required files for the first assessment submission, you will resubmit that file. The new file will overwrite the previous one.

But how many times can you do this? **Maximum three times**.

This means:

1. The first time you make a mistake is okay; you can resubmit.
2. If you realize a second time that maybe a report text file or requirement is missing, you can submit again.
3. You can submit your final submission before the deadline.

4. INTERACTION EXTRACTION (ENGLISH ONLY)

(Note: This segment of the transcript contains primarily instructor monologue for demonstration and navigation. No direct student questions or interactions are present in this

specific text block.)

5. FULL CHINESE TRANSLATION

评估详情与导航

如果你向下滚动，它是针对不同课程的：

- **MSA 800** 用于你的研究方法。
- 其他课程的高性能计算。

评估数量：

- 对于 **MSA 800**（专业软件开发），是 **三个**。
- 其余课程，除了期末考试，有 **两个**。
- 只有 **MSA 800** 以及期末考试，你有三个正式提交。

请不要停止追踪，因为有时你可能会记错。例如，对于量子计算，我们有两个提交，你可能会错过其中一个截止日期。

访问评估信息（MSA 800 示例）

对于 **MSA 800**，第一个评估的详细信息在系统中。我将展示给你看。

在你平台的左上角，有一个名为 **编程课程评估信息** 的部分。

在该页面上，你有评估信息。请点击它。

然后你可以看到 **MSA 800**。有一个总体的详细说明。

关于第一个评估的更多细节，有一个链接。你需要点击那个链接。

你将看到一个 PDF 文件。如果你点击它，你将看到：

- **评估一** 的详细要求。
- 评估背后的场景。
- 提交的 **最低要求**。
- 你需要在截止日期前提交的项目。
- 所有不同部分的要求。

关键任务是 *花时间仔细阅读它*。

评分说明

最后，有一些要求。如果你的目标仅仅是达到 **最低要求**（我不推荐这样做），那对应的就是 **C** 等级。**C 意味着 50 分以上**。你不能低于 50 分。

好的，我们继续。对于所有课程的每个评估，评分标准中 **A、B 或 C** 类别的具体要求都写在那里以供澄清。

阅读需要时间。这就是为什么我现在打开这个，让你在本周早些时候仔细阅读。如果你有任何问题，到时可以问我。

这个评估详情需要你花时间，并且你可能需要重新回顾。我们将讨论并回答你遇到的任何难题。

提交流程与限制

好的，我们正努力使导航更简单，以便你查看提交要素。

有一个提交部分。**你可以提交多少次？** 最多 三 次。是的，三次是截止日期前允许的最大提交次数。

例如，第一个评估的截止日期是 **第七周** 结束。

如果出于任何原因，你错过了第一个评估提交所需的某个文件，你将重新提交该文件。新文件将覆盖之前的文件。

但你可以这样做多少次？**最多三次**。

这意味着：

1. 第一次犯错没关系；你可以重新提交。
2. 如果你第二次意识到可能缺少一个报告文本文件或要求，你可以再次提交。
3. 你可以在截止日期前提交最终版本。

SUBMISSION AND ACADEMIC INTEGRITY



Student: So for every student, is the project the same? Or like... the project or assignment types? Can we choose that?

 **Mohammad:** Okay. Thank you for the question. This is a really good question.

For the first assessment, it is the same assessment for all. It is an **individual** project.

- You need to work on it on your own.
- It's not a group project.
- The first assessment is a solo, individual task. It means one person works and focuses on learning for the development.

 **Mohammad:** Okay, is it clear?

We will not accept copy and paste from each other. We are going to use **Turnitin**.

Turnitin is another platform. I will share the username and password for the course later because you should have access based on your student ID. If you haven't checked it, we will check it later on.

All of the reports (not your code, your *report*) need to be submitted in Turnitin.

- Turnitin will provide a **similarity report** indicating where the content of your report comes from.
- Is it written by you, or did you copy-paste from AI, or did you get it from your friends' work in the past or from business sources?

The similarity report, in the worst case, should be a **maximum of 20 percent**. However, don't aim for 20%.

- If you have genuine work, usually a **5-6 percent similarity** shows up.
- The report will show you where these similarities come from. We will provide the full report.

You need to be patient because, depending on the file size, processing can take time.

- If you prepare one PDF or Word document, the processing time depends on the size.
 - If it's 300 pages, it's totally different. We had 30 pages before; everything will take maybe a minute to analyze and generate the full report.
 - At the beginning, you will see a receipt. We don't count on the receipt, of course.
-

中文全文翻译

提交与学术诚信

🧑 学生：那么对每个学生来说，项目是一样的吗？或者项目或作业的类型？我们可以选择吗？

🧑 **Mohammad**：好的。谢谢你的问题。这是一个非常好的问题。

对于第一个评估，所有学生的评估内容都是相同的。这是一个个人项目。

- 你需要独立完成。
- 这不是一个小组项目。
- 第一个评估是单人、个人的任务。这意味着由一个人独立工作并专注于开发学习。

🧑 **Mohammad**：好的，清楚了吗？

我们不会接受相互之间的复制粘贴。我们将使用 **Turnitin**。

Turnitin 是另一个平台。我稍后会分享课程的登录名和密码，因为你应该能根据学生ID访问。如果你还没有查看，我们稍后会检查。

所有的报告（不是你的代码，是你的报告）都需要在 Turnitin 中提交。

- Turnitin 会提供一份**相似性报告**，指出你报告内容的来源。
- 它是你自己写的，还是你从AI复制粘贴的，或者是从你朋友过去的工作或商业来源获取的？

相似性报告，在最坏的情况下，应**最多为20%**。但是，不要以20%为目标。

- 如果你的工作是原创的，通常会出现**5-6%的相似度**。
- 报告会显示这些相似之处来自哪里。我们将提供完整的报告。

你需要有耐心，因为根据文件大小，处理可能需要时间。

- 如果你准备一个PDF或Word文档，处理时间取决于文件大小。
- 如果是300页，那就完全不同了。我们之前有30页；分析并生成完整报告可能需要大约一分钟。
- 一开始，你会看到一个回执。当然，我们不看回执。

TURNITIN SUBMISSION AND SIMILARITY REPORT

Processing Time and Report Generation

- The processing time for a similarity check depends on the file size.
 - For example, a 30-page document might take about a minute to analyze and generate the full report.
 - A 300-page document would be a completely different matter.
- Initially, you will see a receipt. We do not review the receipt.
- I will have access to all your submissions, including any resubmissions made before the deadline, through Turnitin on Blackboard.

Submission Policy and Similarity Threshold

- **Do not submit** if the similarity report shows **more than 20%**.
- Blackboard will automatically block the submission of a new file with a similarity score above this threshold.
- If your similarity is at 20%, you must reduce it.

How to Reduce Similarity

1. Check the report to identify which parts are flagged as similar.
2. You need to paraphrase those sections.
3. Write them in your own words.
4. Re-check the document through Turnitin.

Resubmission Attempts

- You have **three chances** to check your report and reduce the similarity.
- Once you are satisfied with your report and the similarity score, you can download the full similarity report and submit it along with your other assessment requirements on Blackboard.

 **Student:** Regarding the resubmission, you mentioned there are three times. For example, if the first submission fails, on the second submission, is there a chance that the submitter still receives an A or A+?

 **Lecturer:** I think there is a little bit of confusion between the re-submission for a maximum of 50% (after the deadline) and resubmission before the deadline.

Clarification on Pre-Deadline Resubmissions

Let me clarify with an example:

- I am working on my assessment now, in week six.
- The deadline is week seven.
- During the week between week six and seven, I realize I missed one of the five requirements. The Blackboard submission link is still open.
- I can upload my entire project again for that assessment. **I do not consider this a formal "resubmission"** in the penalized sense.

Key Points on Multiple Uploads Before Deadline

- You have the chance to submit your assessment multiple times before the deadline.
 - It does not matter if it's the first, second, or third upload before the deadline.
 - **I will mark and evaluate only your last submission before the deadline.**
 - If you miss something, you must upload the entire submission again, not just a single file.
 - Blackboard will automatically...
-

TURNITIN 提交与相似性报告

处理时间与报告生成

- 相似性检查的处理时间取决于文件大小。
 - 例如，一份30页的文档可能需要大约一分钟来分析并生成完整报告。
 - 一份300页的文档则完全不同。
- 最初，你会看到一个回执。我们不审查回执。
- 我将能够通过 Blackboard 上的 Turnitin 访问你所有的提交，包括截止日期前的任何重新提交。

提交政策与相似性阈值

- 如果相似性报告显示**超过20%**，**请勿提交**。
- Blackboard 将自动阻止相似性分数超过此阈值的新文件提交。
- 如果你的相似性是20%，你必须降低它。

如何降低相似性

1. 检查报告，找出哪些部分被标记为相似。
2. 你需要改写这些部分。
3. 用你自己的话重写。
4. 通过 Turnitin 重新检查文档。

重新提交尝试

- 你有**三次机会**来检查你的报告并降低相似性。
- 一旦你对报告和相似性分数感到满意，就可以下载完整的相似性报告，并将其与你在 Blackboard 上的其他评估要求一起提交。

 **学生：**关于重新提交，你提到了有三次机会。例如，如果第一次提交失败了，在第二次提交时，提交者是否仍有机会获得A或A+？

 **讲师：**我认为这里对截止日期后最多50%的重新提交和截止日期前的重新提交有点混淆。

关于截止日期前重新提交的澄清

让我用一个例子来澄清：

- 我现在（第六周）正在做我的评估。
- 截止日期是第七周。
- 在第六周和第七周之间，我意识到我漏掉了五个要求中的一个。Blackboard 的提交链接仍然开放。
- 我可以再次上传我的整个项目进行评估。**我不认为这是正式的、会被扣分的"重新提交"。**

关于截止日期前多次上传的要点

- 在截止日期前，你有多次提交评估的机会。
- 无论是第一次、第二次还是第三次上传（在截止日期前）都无关紧要。
- **我将只对截止日期前你的最后一次提交进行评分和评估。**
- 如果你遗漏了某些内容，你必须再次上传整个提交文件，而不是单个文件。
- Blackboard 会自动...

Mohammad: ...automatically show me that you have, for example, two or three submissions.

But which one will be marked? The last one.

Okay? There is no issue. Maybe you're gonna get a B, you're gonna get an A. It depends on the quality of your work.

SCENARIO 2: MISSING THE DEADLINE

But we have another scenario: you miss the deadline.

By the end of Week 7, the element is not accessible for you anymore. After the deadline, you cannot upload any file for Assessment 1.

For your circumstances, we will go to the academic processing. You need to submit a letter explaining why these things happened.

The academic board will make a decision on whether you are eligible to submit or not.

If they make a decision "yes, you can submit," but this submission is for a medical issue or other exceptional cases. We have different scenarios for different cases.

We don't want to waste the time of our class for these exceptional cases. The academic board will make a decision.

SCENARIO 3: BORDERLINE MARKS

But if you submit by the deadline and your mark is, for example, 49... 49 means?

Okay. Then again, the academic board will discuss it.

You will need to be in a meeting with the academic board, of course including your lecturer, to clarify the reason.

Because we don't want to see this situation again in the future. You have this chance just once.

We will not think about getting a fail because I haven't faced this situation with any of my students who are really engaged in the class.

If students just miss the class automatically, yes, because you don't know what the process is, and maybe you're gonna miss a deadline, maybe the quality of your work is too low.

But if you are in the class, follow the class, and study well, there is no issue to get a pass.

Because you're gonna be a software developer. If you're gonna get paid for programming, what are you gonna do?

Because this course is the key for other courses and is also a prerequisite.

What does "prerequisite" mean? It means if you cannot pass this course, you are not eligible to take other courses for the next semester.

THE IMPORTANCE OF A STRONG CODING FOUNDATION

That's why it's called... I didn't teach it before for them three or four semesters ago, but I taught it last semester and at the master's level.

I realized that the students need more support for coding.

Okay, we need a little bit stronger background for coding.

When you are reaching, for example, the next semester for the blockchain, data analytics courses, or AI/ML or whatever, you need to be really confident in your coding.

That's why, okay, even though the first semester is a little bit more challenging, especially for me, but I took that course.

We need to have a stronger background in coding from the students.

That's why I'm with you this semester, to have the Python programming as MSA 800, to make sure your foundation is...

中文全文翻译

Mohammad: ...会自动显示你有，例如，两次或三次提交。

但哪一份会被评分？最后一份。

明白吗？这没有问题。也许你会得B，也许你会得A。这取决于你作业的质量。

情景二：错过截止日期

但我们有另一种情况：你错过了截止日期。

到第7周结束时，该提交入口对你将不再可用。截止日期后，你无法再为评估一上传任何文件。

对于你的个人情况，我们将进入学术处理流程。你需要提交一份说明信，解释事情发生的原因。

学术委员会将决定你是否有资格提交。

如果他们决定"可以，你可以提交"，但这次提交是针对医疗问题或其他特殊情况。针对不同的情况，我们有不同的处理方案。

我们不想在课堂上浪费时间讨论这些特殊情况。学术委员会会做出决定。

情景三：临界分数

但如果你在截止日期前提交，而你的分数是，例如，49分... 49分意味着？

好吧。那么，学术委员会将再次进行讨论。

你将需要参加学术委员会的会议，当然包括你的讲师，以澄清原因。

因为我们不希望未来再看到这种情况。你只有这一次机会。

我们不会考虑让你不及格，因为我还没有遇到过任何真正投入课堂学习的学生出现这种情况。

如果学生只是自动地缺课，是的，因为你不知道流程是什么，也许你会错过截止日期，也许你作业的质量太低。

但如果你在课堂上，跟着课程学习，并且学得很好，通过是没有问题的。

因为你将成为一名软件开发人员。如果你要靠编程获得报酬，你该怎么办？

因为这门课程是其他课程的关键，也是一门先修课程。

"先修课程"是什么意思？这意味着如果你不能通过这门课程，你就不具备在下学期选修其他课程的资格。

强大编码基础的重要性

这就是为什么它被称为... 我几个学期前没有教过他们这门课，但我上学期教过，并且在硕士级别教过。

我意识到学生们在编码方面需要更多的支持。

好吧，我们需要更强一点的编码背景。

当你进入下学期，例如学习区块链、数据分析课程，或人工智能/机器学习等课程时，你需要对自己的编码能力非常有信心。

这就是为什么，好吧，尽管第一学期的课程更具挑战性，尤其对我来说，但我承担了这门课。

我们需要学生具备更强的编码背景。

这就是为什么我这学期和你们在一起，开设MSA 800 Python编程课程，以确保你们的基础...

1. ABSOLUTE FIDELITY PRINCIPLE (ENGLISH ONLY)

Mohammad: Okay, even though the first semester courses are a little bit more challenging, especially for me, but I took that course. We need to have a stronger background in coding from the students.

That's why I'm with you this semester to have the Python programming as MSA 800, to make sure your foundation is quite strong and you are confident for the other courses.

👤 **Mohammad:** Did I answer your question? 👤 **Student:** Yes, yes. 👤 **Mohammad:** Is that clear? 👤 **Student:** Yes. 👤 **Mohammad:** How about your question? 👤 **Student:** Thank you. 👤 **Mohammad:** Perfect.

Mohammad: Any other questions for assessment?

👤 **Student:** Yes. 👤 **Speaker 3:** How could we check our similarity before we submit something?

Mohammad: Okay, again, I'm going to answer. I think you got how we can, but we have another platform. Its name is **Turnitin**.

It's a web application. You have access already. I'm going to share with you the name. You can check it. Don't worry about that.

You're going to use your student email and then you can log in. For each course, you will have a username and a password code. You can go through that course. That code will be shared by the lecturer during the semester.

For example:

- For MSA 800, I will share it.
- For Research Method, I think RIM will share with you a quantum arrow.

Then you can log in for MSA 800 and go through. There is a section where you can browse and find a file that you want to submit for Assessment 1, for example, the PDF file (your report). And the system will generate a similarity report of your document.

If you are not happy because you want to see, it will be highlighted which part is copied from somewhere and where it's gone from. Is it generated by AI or is it generated by you?

I recommend using AI in the right way. I'm not asking not to use Gemini, like PC, I don't know, Claude, many more, ChatGPT. You can use it, but don't leave the thinking on behalf of you. These tools are to assist you.

Like, I don't know, 20, 30 years ago, Google search engine was new. Okay, we're gonna get an answer. It was something like LLM models now. But after a while, we realized that they are not the solution. They are just helping us to find the best answer and save time to be efficient.

If you're going to copy and paste, and then you don't know what is going on in your report, and I ask you to present it...

Because the second assessment is a group, is a team work. It is two people maximum and minimum. We would prefer all of you being in groups of two. If we have one left as a solo, it's fine. I cannot support, but I recommend.

On the second assessment, we will not talk about now, it's too soon. It is a group project, and the project will be defined by you and approved by me. If the scope of the project is good enough, the project that you're gonna define, maybe I'm gonna change, add more features...

5. 中文全文翻译

Mohammad: 好的，尽管第一学期的课程更具挑战性，尤其对我来说，但我承担了这门课。我们需要学生具备更强的编码背景。

这就是为什么我这学期和你们在一起，开设MSA 800 Python编程课程，以确保你们的基础相当扎实，并对其他课程充满信心。

🧑🏫 **Mohammad:** 我回答你的问题了吗？ 🧑🏫 **学生:** 是的，是的。 🧑🏫 **Mohammad:** 清楚了吗？ 🧑🏫 **学生:** 是的。 🧑🏫 **Mohammad:** 你的问题呢？ 🧑🏫 **学生:** 谢谢。 🧑🏫 **Mohammad:** 很好。

Mohammad: 关于考核还有其他问题吗？

🧑🏫 **学生:** 有。 🧑🏫 **Speaker 3:** 我们在提交作业前如何检查相似度？

Mohammad: 好的，我再回答一次。我想你们已经知道一种方法，但我们还有另一个平台。它的名字叫 **Turnitin**。

它是一个网络应用程序。你们已经可以访问了。我会把名称分享给你们。你们可以查看。不用担心这个。

你们需要使用学生邮箱登录。对于每门课程，你们会有一个用户名和一个密码代码。你们可以进入那门课程。该代码将在学期内由授课讲师分享。

例如：

- 对于MSA 800，我会分享代码。
- 对于研究方法论，我想RIM会分享给你们一个量子箭头。

然后你们就可以登录MSA 800并进入。有一个版块，你们可以浏览并找到想要提交的文件，例如用于评估1的PDF文件（你们的报告）。系统会生成你们文档的相似度报告。

如果你们不满意，因为你们想查看，它会高亮显示哪部分是从某处复制的，以及来源是哪里。它是AI生成的还是你们自己生成的？

我建议以正确的方式使用AI。我不是要求不使用Gemini，比如PC，我不知道，Claude，还有很多，ChatGPT。你们可以使用它，但不要让它代替你们思考。这些工具是用来辅助你们的。

就像，我不知道，20、30年前，谷歌搜索引擎是新生事物。好吧，我们会得到一个答案。它有点像现在的LLM模型。但过了一段时间，我们意识到它们不是解决方案。它们只是帮助我们找到最佳答案，并节省时间以提高效率。

如果你们只是复制粘贴，然后你们自己都不知道报告里写了什么，而我要求你们进行演示.....

因为第二次评估是小组作业，是团队合作。最多两人，最少也是两人。我们更希望你们所有人都组成两人小组。如果最后剩下一个人单独一组，也可以。我无法提供支持，但我建议如此。

关于第二次评估，我们现在先不谈，为时过早。它是一个小组项目，项目将由你们定义并由我批准。如果项目范围足够好，你们定义的项目，我可能会进行修改，增加更多功能.....

ASSESSMENT 2: GROUP PROJECT DETAILS

We would prefer all of you to be in groups of two. If we have one person left as a solo, it is fine. I cannot provide support, but I recommend it.

We will not talk about the second assessment now; it is too soon. It is a group project, and the project will be defined by you and approved by me.

If the scope of the project you define is good enough, I might change it:

- I may add more features.
- Or, I may remove some features if I think it is too much for you.

During the six or seven weeks you have time, you will work with your team member for the second assessment.

 **Student:** If you are not happy with your team member...  **Lecturer:** I don't care. You know why? Because in the future, you are going to work in the industry with your colleagues. Can you choose? "I don't want this guy, I don't want this lady to be my colleague." No, you are going to work in the company, and a couple of them have been working on that project for a long time, and you are new. But even someone newer than you is going to join, and you are not... You need to improve your **soft skills** to be able to communicate with everyone.

That's why you can choose your team member. Okay, it's fair. But when you choose, you need to work in the middle of the project. Of course, you are going to face lots of challenges. These challenges will help you to grow, not just run away. "All I don't want this, I'm gonna do it by myself."

We had these kinds of scenarios, I think two or three semesters back, but we didn't have it last semester because I explained: you are going to choose the team member. You need to be really wise to choose whoever you like to work together. It doesn't mean you are going to

be friends forever, but you are going to work professionally together to deliver the project in the best way.

Because if the project you are going to submit is good, the positive consequences will be for both of you. The same as in industry: when you are a team of five and work on a project, and the outcome is quite convincing for the end user or a stakeholder, then the bonus will be for all of those, including the company and many more people. But if it is a failed project, the consequence is again for all of you. You cannot tell, "Oh, I did my best, I did my part."

Another thing: it's not a way to complain. We need to get this mindset of what we are doing in the team. If your team member is a little bit behind, you need to support. Don't leave it, "Oh, okay, he's not doing the UI, I am just doing the backend and the project." Okay, up to you. How are you going to manage? Because you are going to manage your team as well.

So it's too soon to talk, but I just wanted to give a heads up.

ASSESSMENT 3: FINAL PRESENTATION

For the third assessment, you are going to be here in week 15, both of you in the group, and present what you have done.

- What is your responsibility?
- Which part is your contribution?

Even though I can see your contribution through GitHub (because you have the joint GitHub), I can see the history of your work, the commits, and all of the challenges that you face and the co...

评估 2：小组项目详情

我们更希望你们所有人都组成两人小组。如果最后剩下一个人单独一组，也可以。我无法提供支持，但我建议如此。

我们现在先不谈第二次评估，为时过早。它是一个小组项目，项目将由你们定义并由我批准。

如果你们定义的项目范围足够好，我可能会进行修改：

- 我可能会增加更多功能。
- 或者，如果我认为对你们来说任务太重，我可能会移除一些功能。

在你们拥有的六到七周时间里，你们将与团队成员一起完成第二次评估。

 **学生：** 如果你对团队成员不满意...  **讲师：** 我不在乎。你知道为什么吗？因为在未来，你将在行业中与你的同事一起工作。你能选择吗？"我不想要这个家伙，我不想要这位女士做我的同事。" 不，你将进入公司工作，他们中有几个人在那个项目上工作了很长时间，而你是新人。但即使比你更新的人也会加入，而你并不是... 你需要提升你的 **软技能**，以便能够与每个人沟通。

这就是为什么你们可以选择自己的团队成员。好吧，这很公平。但是当你们选择后，就需要在项目中合作。当然，你们会遇到很多挑战。这些挑战将帮助你们成长，而不是逃避。"我什么都不想要，我要自己一个人做。"

我们以前遇到过这种情况，我想是两三个学期前，但上学期没有，因为我解释过：你们将选择团队成员。你们需要非常明智地选择你们愿意一起工作的人。这并不意味着你们要成为永远的朋友，而是要专业地合作，以最佳方式交付项目。

因为如果你们将要提交的项目是好的，积极的成果将惠及你们双方。这与行业中的情况相同：当你们是一个五人团队，共同完成一个项目，并且成果对最终用户或利益相关者非常有说服力时，那么奖金将属于所有人，包括公司和更多相关人员。但如果是一个失败的项目，后果同样由你们所有人承担。你不能说："哦，我尽力了，我完成了我的部分。"

另一件事：这不是抱怨的方式。我们需要建立这种团队合作的心态。如果你的团队成员稍微落后了，你需要提供支持。不要放任不管，"哦，好吧，他不~~做~~UI，我只做后端和项目。" 好吧，随你便。你打算如何管理？因为你也需要管理你的团队。

所以现在谈论这个还为时过早，但我只是想提前告知一下。

评估 3：最终演示

对于第三次评估，你们将在第15周来到这里，小组的两人~~都要~~到场，并演示你们完成的工作。

- 你的职责是什么？
- 哪部分是你的贡献？

即使我可以通过 GitHub 看到你们的贡献（因为你们有共享的 GitHub），我可以看到你们工作的历史记录、提交记录以及你们面临的所有挑战和协...

ASSESSMENT 3: FINAL PRESENTATION (CONTINUED)

...作，以及你们留下的评论。

我建议你们在第一次评估时就使用 **GitHub**，即使它是一个个人项目，以便为第二次评估（小组项目）做好准备。

但是，第三次评估（最终演示）是一个重要的评估。你们需要在这里展示，你们的工作质量将说明一切。

- 如果团队中一个人只是坐在一边写报告，而另一个人负责编码，这对你们的团队不利。
- 这意味着你们两人都会受到负面影响。

如果你们的团队成员需要分担责任，那么每个人都应该承担文档和编码的责任。

- 如果一个人负责整个项目的编码部分，而他的团队成员只是享受生活，这是不公平的。
- 因为在未来，你们会遇到很多困难。

我想我之前举过这个例子：

 **Student:** Am I right?  **Mohammad:** If you want to be a good driver, you need to sit and drive. If you want to be a good developer, you cannot sit next to the driver and just read the map. "Oh, okay, go to the left, go to the right. And no, you made a mistake here." We don't want that.

我们希望你们都能成为优秀的"驾驶员"，并互相支持。

好了，关于评估就讲到这里。

课程资源与期望

我想现在最好通过 **LMS** 平台来了解你们可以在哪里找到课程内容和资源。

- 如果你们需要更多，请向我提问。
- 如果你们问得更多，并且我能看到你们的热情，我会分享更多、更多、更多。
- 如果没有，我将只维持标准水平的内容。

这取决于你们的热情。我非常乐意分享更多，但我需要看到你们能够运用我分享的内容。如果我分享了一些非常高级的东西，而你们没有任何头绪，那就没有意义了。

课堂活动：环境检查与项目规范

好了，我喜欢你们的热情。现在我们需要在课堂上进行一项活动，基于第一周的内容进行回顾。

因为第一周是在线进行的，我想确保你们每个人的软件设置都没问题，并且你们清楚自己做什么。

 **Student:** Can I develop a project without an IDE? Of course, you can do scripting. You know, one piece of code and running it through the terminal. Can I use Jupyter Notebook for just testing?  **Mohammad:** Yes,对于任何测试活动，你可以使用 Jupyter Notebook。但是，对于我将要分享给你们的任何活动，都应该是完整的项目。

这意味着你们需要将其作为一个 **Python** 项目来编写，通过 **VS Code** 或 **PyCharm** 或类似的 **IDE** 来完成。

我不会仅仅依赖 **Jupyter Notebooks**，但你们可以将其用于测试。我们以后会更多地使用它，但我不希望你们只习惯使用 Jupyter Notebooks。

因为在未来的工业界，他们不会仅仅接受你只熟悉和使用那个环境。我们应该使用 **VS Code** 或其他 **IDE**。

你们总是有时间和许可去尝试、犯错、休息一下。

评估 3：最终演示（续）

...作，以及你们留下的评论。

我建议你们在第一次评估时就使用 **GitHub**，即使它是一个个人项目，以便为第二次评估（小组项目）做好准备。

但是，第三次评估（最终演示）是一个重要的评估。你们需要在这里展示，你们的工作质量将说明一切。

- 如果团队中一个人只是坐在一边写报告，而另一个人负责编码，这对你们的团队不利。
- 这意味着你们两人都会受到负面影响。

如果你们的团队成员需要分担责任，那么每个人都应该承担文档和编码的责任。

- 如果一个人负责整个项目的编码部分，而他的团队成员只是享受生活，这是不公平的。
- 因为在未来，你们会遇到很多困难。

我想我之前举过这个例子：

🧑‍🎓 学生：我说得对吗？🧑‍🎓 **Mohammad**：如果你想成为一名好司机，你需要坐上去驾驶。如果你想成为一名优秀的开发者，你不能只是坐在司机旁边看地图。"哦，好的，向左，向右。不对，你这里搞错了。"我们不希望那样。

我们希望你们都能成为优秀的"驾驶员"，并互相支持。

好了，关于评估就讲到这里。

课程资源与期望

我想现在最好通过 **LMS** 平台来了解你们可以在哪里找到课程内容和资源。

- 如果你们需要更多，请向我提问。
- 如果你们问得更多，并且我能看到你们的热情，我会分享更多、更多、更多。
- 如果没有，我将只维持标准水平的内容。

这取决于你们的热情。我非常乐意分享更多，但我需要看到你们能够运用我分享的内容。如果我分享了一些非常高级的东西，而你们没有任何头绪，那就没有意义了。

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🧑‍🎓 学生：我可以在没有 IDE 的情况下开发项目吗？当然，你可以写脚本。你知道的，一段代码然后在终端里运行。我可以用 Jupyter Notebook 只做测试吗？🧑‍🎓 **Mohammad**：是的，对于任何测试活动，你可以使用 Jupyter Notebook。但是，对于我将要分享给你们的任何活动，都应该是完整的项目。

这意味着你们需要将其作为一个 **Python** 项目来编写，通过 **VS Code** 或 **PyCharm** 或类似的 **IDE** 来完成。

我不会仅仅依赖 **Jupyter Notebooks**，但你们可以将其用于测试。我们以后会更多地使用它，但我不希望你们只习惯使用 Jupyter Notebooks。

因为在未来的工业界，他们不会仅仅接受你只熟悉和使用那个环境。我们应该使用 **VS Code** 或其他 **IDE**。

你们总是有时间和许可去尝试、犯错、休息一下。

Professional Software Engineering Course - Session Transcript (Cont'd)

Classroom Culture & Expectations

ON USING JUPYTER NOTEBOOKS

I will not rely solely on **Jupyter Notebooks**, but you can use it for testing. We will use it more later on, but I don't want you to get used to using *just* Jupyter Notebooks.

Because in the future industry, they will not accept that you are only confident and have experience using that environment. We should use **VS Code** or any other **IDE**.

ON TIME MANAGEMENT AND CONDUCT

You always have the time and permission to try, make mistakes, and take a break for recharging. You don't need to be perfect. At least in my class, you are quite open. We can go out and come back, but be wise. All of us are mature enough to consider the time.

ON ASKING QUESTIONS

Another thing: there is **no dumb question** in our class.

It means if the question is quite repetitive, it's totally fine. I will answer it, or sometimes I will ask your classmates to support you in answering, because maybe you missed that point.

It also means you are **not allowed to complain** if your friend asks any questions, from simple to really high-level. You need to respect each other because the class is for all of us.

Course Activities & Structure

ACTIVITY NUMBERING SYSTEM

Now, we're going to go to our activities. I will put a number on all of the activities. Then it will be easier for us because later on, we will go back for activity number 2 in week 1, and then we want to expand.

 **Speaker 3:** Then it's easier for you to find.

NATURE OF ACTIVITIES

All of the activities in the class are quite new. It means based on the progress in our class, I will write new activities. The activity does not come from last semester or something that is predefined.

When we are discussing, even for a new project, I will share something based on your questions. That's why sometimes you need to be patient for one or two minutes so I can share that activity.

Activity 1: Project Clarification

PROJECT REQUIREMENTS

 **Speaker 2:** Activity on your team, please check activity number one.

 **Mohammad:** Is this a simple project just as a backend? We need to calculate using the **complex data type**.

Because I mentioned the program should support operators with the complex and other data. It means we want to use complex and maybe integer, float, or whatever that you need.

MINIMUM FUNCTIONALITY

You need a minimum of three functionalities—sorry, three operators. So you can use plus, minus...

👤 **Mohammad**: See here... Okay.

👤 **Mohammad**: How many functions do we need to have? Minimal three functions in your project. Of course, you need `main` as well. Okay?

So I h...

专业软件工程课程 - 会话记录（续）

课堂文化与期望

关于使用 JUPYTER NOTEBOOKS

我不会仅仅依赖 **Jupyter Notebooks**，但你们可以将其用于测试。我们以后会更多地使用它，但我不希望你们只习惯使用 仅仅 Jupyter Notebooks。

因为在未来的工业界，他们不会仅仅接受你只熟悉和使用那个环境。我们应该使用 **VS Code** 或其他 **IDE**。

关于时间管理与行为规范

你们总是有时间和许可去尝试、犯错、休息一下来充电。你们不需要做到完美。至少在我的课堂上，你们是相当开放的。我们可以出去再回来，但要明智。我们所有人都足够成熟，会考虑到时间。

关于提问

另一件事：在我们的课堂上，**没有愚蠢的问题**。

这意味着如果问题相当重复，完全没问题。我会回答，或者有时我会请你的同学来帮助你回答，因为你可能错过了那个点。

这也意味着，如果你的朋友提出任何问题，从简单的到非常高阶的，你**不允许抱怨**。你们需要互相尊重，因为课堂是属于我们所有人的。

课程活动与结构

活动编号系统

现在，我们要开始我们的活动。我会给所有活动编号。这样对我们来说会更容易，因为以后我们会回到第1周的活动2，然后我们想进行扩展。

 **Speaker 3:** 这样你更容易找到。

活动的性质

课堂上的所有活动都是相当新的。这意味着根据我们课堂的进度，我会编写新的活动。这些活动不是来自上学期或预先定义好的内容。

当我们讨论时，即使是一个新项目，我也会根据你们的问题来分享一些内容。这就是为什么有时你们需要耐心等待一两分钟，以便我能分享那个活动。

活动 1：项目澄清

项目要求

 **Speaker 2:** 关于你们团队的活动，请查看活动一。

 **Mohammad:** 这是一个简单的仅作为后端的项目吗？我们需要使用复数数据类型进行计算。

因为我提到过，程序应该支持对复数和其他数据类型的运算符。这意味着我们想使用复数，也许还有整数、浮点数，或者任何你们需要的数据类型。

最低功能要求

你们需要至少三个功能——抱歉，是三个运算符。所以你们可以使用加、减...

 **Mohammad:** 看这里... 好的。

 **Mohammad:** 我们需要有多少个函数？在你的项目中至少三个函数。当然，你们也需要 `main` 函数。好吗？

所以我...

ACTIVITY CLARIFICATION AND INSTRUCTIONS

Mohammad: You need to use complex numbers, and maybe integers, floats, or whatever data type you need.

The **minimum** requirement is three functionalities — sorry, three operators. So you can use plus, minus...

 **Mohammad:** See here... Okay.

 **Mohammad:** How many functions do we need to have? At least three functions in your project. Of course, you need a `main` function as well. Okay?

So I have the mic. Any questions about this activity? Is any part unclear?

 **Speaker 2:** No.

 **Mohammad:** All good? How many minutes do you need to do it? 10 to 15 minutes for now? The whole scenario. Maybe you can finish it faster.

IMPORTANT GUIDANCE ON CODE SUBMISSION

Any challenges from decoding come from copy-pasting from ChatGPT or somewhere else. You know why?

Because if you want to share something with me, I'm going to ask you: "Okay, what is this syntax? How do we use this one?"

I know the result might be good and correct, but the problem is you're learning. You need to understand whatever you're going to share. That's all that matters.

It doesn't matter what your current coding level is. If you write in a simple way but you understand it, that's okay. Then we will learn more and more in the best way.

Don't be overly conservative. I need to know if you're okay. That's why, if you don't know anything, please ask.

For example, if you're going to use a complex data variable as a simple operator — for instance, for `plus`, we have two variables as complex numbers — then you need to know how these two complex variables will, for example, multiply or add with each other.

If you don't know, please do a little bit of research. I will show you.

I will put up a timer to see how long it takes for all of you to share your results.

 **Mohammad:** I'm putting the timer on now, as if it's already started. Let's see how it goes.

SESSION CONCLUSION AND STUDENT INTERACTIONS

中文全文翻译

活动澄清与说明

Mohammad: 你们需要使用复数，也许还有整数、浮点数，或者任何你们需要的数据类型。

最低要求是三个功能——抱歉，是三个运算符。所以你们可以使用加、减...

 **Mohammad:** 看这里... 好的。

 **Mohammad:** 我们需要有多少个函数？在你的项目中至少三个函数。当然，你们也需要一个 `main` 函数。好吗？

我这里有麦克风。对这个活动有什么问题吗？有任何不清楚的部分吗？

 **学生 2:** 没有。

 **Mohammad:** 都清楚了吗？你们需要多少分钟来完成？现在给10到15分钟？整个场景。也许你们可以完成得更快。

关于代码提交的重要指导

任何解码的挑战都来自于从ChatGPT或其他地方复制粘贴。你们知道为什么吗？

因为如果你们想和我分享一些东西，我会问你们："好的，这个语法是什么？我们怎么使用这个？"

我知道结果可能是好的、正确的，但问题在于你们正在学习。你们需要理解你们将要分享的任何内容。这才是最重要的。

你们目前的编码水平如何并不重要。如果你们用简单的方式写但能理解，那也没关系。然后我们会以最好的方式学习更多。

不要过于保守。我需要知道你们是否没问题。这就是为什么，如果你们有任何不懂的，请提问。

例如，如果你们打算使用一个复数数据变量作为一个简单的运算符——比如，对于 **加** 运算，我们有两个复数变量——那么你们需要知道这两个复数变量将如何，例如，相乘或相加。

如果你们不知道，请做一点研究。我会演示给你们看。

我会设置一个计时器，看看你们所有人分享结果需要多长时间。

 **Mohammad:** 我现在开始计时，就当它已经开始了。让我们看看进展如何。

课程结束与学生互动

Mohammad: Maybe 10 more minutes. **Mohammad:** Okay? **Mohammad:** Because you haven't finished it. **Mohammad:** Let's give you a little bit longer time. **Mohammad:** I think it's about plus 15 minutes here.

Speaker 2: It's okay.

Mohammad: I'm not sure how you report checking from 10:15. **Mohammad:** Okay, just keep working on your activity. **Mohammad:** I will share something on your themes after finishing your activity. **Mohammad:** You can learn how we can connect to the 10:15. **Mohammad:** Okay. **Mohammad:** Thank you. **Mohammad:** Thank you. **Mohammad:** Okay. **Mohammad:** Thank you. **Mohammad:** yeah

Speaker 2: 他只要把他船上 **Speaker 2:** 可是他刚才说 **Speaker 2:** 对 **Speaker 3:** 你已经搞定了 **Speaker 3:** 你允许你解我什么 **Speaker 4:** 他典型的一些计算方式 **Speaker 4:** 我问的结果就是 **Speaker 4:** 你所有的一些计算 **Speaker 4:** 然后让他跟你算出来 **Speaker 3:** 对 **Speaker 3:** 所以这个明显是错的 **Speaker 3:** 他就是把那个战事来 **Speaker 4:** 所以你要改,我之前,我刚才申请的第一版也是,然后就要改,也就是说,云行的时候你需要输入一些, **Speaker 4:** 嗯,input,一些parameter,然后,然后这个,这个,这个,这个,然后再拿parad计刷,然后它跟你算句结果

Mohammad: Can you see my text on the themes for the penalty? **Mohammad:** yes or no okay for the eternity I put a link also the code for this course **Mohammad:** please your use your student email you be a student email you can log in through **Mohammad:** your eternity I cannot show here **Mohammad:** There is a login page as a student or lecturer and also admin. **Mohammad:** to go for the student login and then you're gonna see this page. If it's the first time, **Mohammad:** possibly you need to reset your password and then you will get a reset option through your **Mohammad:** email address. Then you can reset your password

and for the first time login. When you login, **Mohammad**: definitely you need a code after login because you have one password and then you need a code **Mohammad**: every course and I share with you on the thing you can use that one and then you **Mohammad**: have something your report in check the similarity because the similarity **Mohammad**: report come from these two needs to be submitted along with other parts for your **Mohammad**: access. Please try to log in because later on I can see how many of you if I **Mohammad**: refresh I can see how many of you already joined the class or how many **Mohammad**: times that you saw me and what is the similarity report in detail. A little bit **Mohammad**: more detail of the report that you can see **Mohammad**: Thank you. **Mohammad**: はい

Speaker 2: Thank you.

Mohammad: I can see 10 of you joined so far, but we need more.

Mohammad: 也许再给10分钟。 **Mohammad**: 好吗？ **Mohammad**: 因为你们还没完成。

Mohammad: 让我们再多给你们一点时间。 **Mohammad**: 我想这里大概再加15分钟。

Speaker 2: 没关系。

Mohammad: 我不确定你们如何从10:15开始报告检查。 **Mohammad**: 好的，请继续完成你们的课堂活动。 **Mohammad**: 在你们完成活动后，我会在你们的主题上分享一些东西。

Mohammad: 你们可以学习我们如何连接到10:15。 **Mohammad**: 好的。 **Mohammad**: 谢谢。 **Mohammad**: 谢谢。 **Mohammad**: 好的。 **Mohammad**: 谢谢。 **Mohammad**: 是的

Speaker 2: 他只要把他船上 **Speaker 2**: 可是他刚才说 **Speaker 2**: 对 **Speaker 3**: 你已经搞定了 **Speaker 3**: 你允许你解我什么 **Speaker 4**: 他典型的一些计算方式 **Speaker 4**: 我问的结果就是 **Speaker 4**: 你所有的一些计算 **Speaker 4**: 然后让他跟你算出来 **Speaker 3**: 对 **Speaker 3**: 所以这个明显是错的 **Speaker 3**: 他就是把那个战事来 **Speaker 4**: 所以你要改, 我之前,我刚才申请的第一版也是,然后就要改,也就是说,云行的时候你需要输入一些, **Speaker 4**: 嗯,input,一些parameter,然后,然后这个,这个,这个,这个,然后再拿parad计刷,然后它跟你算句结果

Mohammad: 你们能看到我在主题上关于惩罚的文本吗？ **Mohammad**: 是或否，好的，为了永恒，我放了一个链接，还有这门课的代码。 **Mohammad**: 请使用你的学生邮箱，你必须是学生邮箱，你可以通过它登录。 **Mohammad**: 你的永恒，我无法在这里展示。 **Mohammad**: 有一个登录页面，作为学生、讲师或管理员。 **Mohammad**: 进入学生登录，然后你会看到这个页面。如果是第一次， **Mohammad**: 你可能需要重置密码，然后你会通过你的邮箱地址收到一个重置选项。 **Mohammad**: 然后你可以重置密码并首次登录。当你登录时， **Mohammad**: 登录后你肯定需要一个代码，因为你有一个密码，然后你每门课程都需要一个代码。 **Mohammad**: 我在那个东西上分享给你们，你们可以使用那个，然后你们 **Mohammad**: 会有一些东西，你的报告在检查相似度，因为相似度 **Mohammad**: 报告来自这

两个，需要与你的访问的其他部分一起提交。 **Mohammad**: 请尝试登录，因为稍后我可以看到你们有多少人，如果我 **Mohammad**: 刷新，我可以看到你们中已经有多少人加入了课堂，或者你们 **Mohammad**: 看了我多少次，以及相似度报告的详细信息。一点 **Mohammad**: 更详细的报告，你们可以看到。 **Mohammad**: 谢谢。 **Mohammad**: 是的

Speaker 2: 谢谢。

Mohammad: 我看到目前有10位同学加入了，但我们需要更多人。

Mohammad: I can see 10 of you joined so far, but we need more.

No one submitting is okay, but 10 students tried to use that code I shared on Teams.

How many are we? I think 38?

35.

35?

35, yeah.

Okay.

But I think we used to be 35.

We're gonna be 38 soon.

But we are already 38.

And maybe we're gonna be 40 soon, but by the way, to train more or less.

Speaker 3: So, there was. **Speaker 2**: All welcome.

Mohammad: Any issues with access?

So far 12.

If you can have access, you can help your friends.

And just to announce, we're going to have a campus tour together.

Are you listening?

At 10:00, we're gonna go together around the campus to get familiar with the fire situation, just in case.

We're gonna take the stairs. I believe you know where the bathroom is.

If you don't know, we're gonna take the stairs. I'm gonna show it to you for the emergency.

As you can see over there, we can see it is the way that for the emergency situation these can be used.

That's why you should use this area for the emergency situation.

Please.

And also we have other things.

During any emergency situation, you shouldn't use the lift.

We should use the steps.

And then we will go down.

We will not go out just randomly; we're gonna check your application to make sure it works properly.

You can use the entrance, open the gate.

Some of you, maybe you are not using the app properly, you need to make sure the Bluetooth is on in your app.

And also, the application should be open to try to have access.

But I think we do, we recognize your cell phone to have access.

Difficulty for the entry for now so far, as if I refresh it I can see possibly more.

Speaker 2: Yeah, 16. **Speaker 2:** Half of the class almost.

Mohammad: Yes, if it's the first time you need to reset it.

Did you reset your password or you used the UBA password?

For the first time.

What did you do?

That's why you can reset and you're gonna get a reset code through your email.

Who logged in?

Who logged in? Did you reset your password?

Create a new user and then based on the UB...

Okay, create a new user and just use the UB email. On the first day you can go to...

Because they already updated the instruction.

Speaker 2: Each semester we have a new way.

Mohammad: What's going on?

OK.

And you have this one?

OK.

中文全文翻译

Mohammad: 目前我看到有10位同学加入了，但我们需要更多人。

没有人提交也没关系，但有10位同学尝试使用了我在Teams上分享的代码。

我们有多少人？我想是38？

35。

35?

35，是的。

好的。

但我想我们以前是35人。

我们很快就会是38人。

但我们已经38人了。

也许我们很快就会是40人，但顺便说一下，大概训练这么多人。

Speaker 3: 所以，有... **Speaker 2:** 欢迎所有人。

Mohammad: 访问有什么问题吗？

目前12人。

如果你能访问，你可以帮助你的朋友。

另外宣布一下，我们将一起进行校园参观。

你们在听吗？

10点钟，我们将一起在校园里走走，熟悉一下火灾情况，以防万一。

我们将走楼梯。我相信你们知道洗手间在哪里。

如果你们不知道，我们将走楼梯。我会指给你们看应急通道。

正如你们在那边看到的，我们可以看到这是紧急情况下可以使用的通道。

这就是为什么你们在紧急情况下应该使用这个区域。

请。

我们还有其他事情。

在任何紧急情况下，你们不应该使用电梯。

我们应该使用楼梯。

然后我们会下去。

我们不会随便出去；我们将检查你们的应用程序以确保其正常工作。

你们可以使用入口，打开门。

你们中的一些人，可能没有正确使用应用程序，你们需要确保应用程序中的蓝牙是打开的。

而且，应用程序应该处于打开状态以尝试访问。

但我想我们确实能识别你们的手机以允许访问。

目前进入有些困难，如果我刷新一下，可能会看到更多人。

Speaker 2: 是的，16人。 **Speaker 2:** 差不多一半的同学了。

Mohammad: 是的，如果是第一次使用，你需要重置它。

你重置了密码还是使用了UBA密码？

第一次使用。

你做了什么？

这就是为什么你可以重置，然后你会通过电子邮件收到一个重置代码。

谁登录了？

谁登录了？你重置密码了吗？

创建一个新用户，然后基于UB...

好的，创建一个新用户，只需使用UB邮箱。第一天你可以去...

因为他们已经更新了说明。

Speaker 2: 每个学期我们都有新方法。

Mohammad: 怎么回事？

好的。

你有这个吗？

好的。

Professional Software Engineering Course Transcript

Course Platform & Repository Setup

Mohammad: Create a new user and then based on the UB... Okay, create a new user and just use the UB email. On the first day you can go to... because they already updated the instruction.

Speaker 2: Each semester we have a new way.

Mohammad: What's going on? OK. And you have this one? OK, click on that. OK. OK. Here is the code that I shared with you on the page for the new course. Is it based on your... you should see all of them. I'm sorry. You need to use this one, the things that I share. Okay? You need the code that I share on the things.

Code Review & Collaborative Learning

Mohammad: I'm going to randomly check one of the answers from your friends... this repository. Anyone? Because sometimes no one will claim the repository that is shared. This one is from your... Is it okay to check your... let's double check and learn with each other.

Mohammad: What is the reason that you add this package? Can you please tell? To do the factorial and other operations. Okay. Because possibly you've got to call the function from this package. That's why you need to import it at the top line. If you want to write the function by yourself, definitely you don't need it. For example, if the factorial... just because like these,

then you have to import it. But you can write a function as well, then you don't need. Let's see how many functions are available.

Mohammad: Okay, I'll create a `B` before the... blocks my name. We have another function for remainder. Okay, please tell me why we use `//` (double slash) next to each other? To remove the absolute decimal part. I just wanted like, what's the number that divides it and then the remainder is... what's the remainder after dividing from that number. Oh, it means if you're gonna use `/` (single slash), then point something. So you need to check what is the difference between one slash and double slash for the division. If you're using remainder, I think it's clear. For example, remainder of five divided by two should be... `5 // 2` should be `2` and `5 % 2` (remaining) should be `1`. Okay? If `a` is `5` and `b` is `2`, the outcome should be `1`.

Mohammad: And then we have another function for the factorial. That's where you can see `math` is important because this function is defined in the library. Its name is `math.factorial`. Did you install it? Because you are using the Anaconda and this pre-installed package. The `math` is a pre-installed package, you don't need to... If installing some targets that you need to install and it's not available, later on we're gonna see. And also he defined the `maximum` is another function. If you use `maximum` as a function or not... the `max(a, b)` is predefined just...

专业软件工程课程字幕

课程平台与仓库设置

Mohammad: 创建一个新用户，然后基于UB... 好的，创建一个新用户，只需使用UB邮箱。第一天你可以去...因为他们已经更新了说明。

Speaker 2: 每个学期我们都有新方法。

Mohammad: 怎么回事？好的。你有这个吗？好的，点击那个。好的。好的。这是我为新课程页面分享给你们的代码。是基于你的...你应该能看到全部。抱歉。你需要用这个，我分享的东西。好吗？你需要我在那些东西上分享的代码。

代码审查与协作学习

Mohammad: 我要随机检查你们一位朋友的答案...这个仓库。有人吗？因为有时候没人会认领分享的这个仓库。这个来自你的...检查你的...可以吗？让我们互相检查，互相学习。

Mohammad: 你添加这个包的原因是什么？能请你告诉我吗？为了做阶乘和其他运算。好的。因为可能你需要从这个包中调用函数。这就是为什么你需要在顶部导入它。如果你想自己写这个函数，那你肯定不需要它。例如，如果阶乘...只是因为像这些，那么你就必须导入它。但你自己也可以写一个函数，那样就不需要了。让我们看看有多少函数可用。

Mohammad: 好的，我将在...之前创建一个 `B` ...挡住我的名字。我们还有另一个求余数的函数。好的，请告诉我为什么我们使用 `//`（双斜杠）？为了移除绝对小数部分。我只是想知道，什么数能整除它，然后余数是...除以那个数后的余数是什么。哦，这意味着如果你使用 `/`（单斜杠），那么会得到小数。所以你需要检查单斜杠和双斜杠在除法上的区别。如果你使用余数，我想很清楚。例如，5除以2的余数应该是... `5 // 2` 应该是 2，而 `5 % 2`（余数）应该是 1。好吗？如果 `a` 是 5，`b` 是 2，结果应该是 1。

Mohammad: 然后我们还有另一个阶乘函数。这就是为什么 `math` 很重要，因为这个函数定义在库中。它的名字是 `math.factorial`。你安装了吗？因为你使用的是Anaconda，这个预安装的包。`math` 是一个预安装的包，你不需要...如果需要安装一些目标包但不可用，我们稍后会看到。他还定义了 `maximum` 是另一个函数。你是否将 `maximum` 用作函数... `max(a, b)` 是预定义的，只是...

1. ABSOLUTE FIDELITY (ENGLISH)

Mohammad: Did you install it? Because you are using Anaconda, and this is a pre-installed package. `math` is a pre-installed package; you don't need to install it. If you need to install some target packages that are not available, we'll see that later on. He also defined `maximum` as another function. Whether you use `maximum` as a function or not... `max(a, b)` is predefined; you just need to use it. Okay, type conversion. You give it permission: if it's an integer, it will convert it to a float. If it's a float, leave it as it is. That's why we have this type conversion here, for two inputs, A and B. Okay, we're going to have these as a complex, `z1` and `z2`, and we're going to return... If you check this part, the result was correct for the complex. How do you print out the outcome of `add_complex`? Is there any main function? Is there any main function in your project? Yes, the compiler... If you are writing just a script, it's okay. But if you are writing a full project, companies look for the `main` function. We should add `main`; that is one part you can improve. Also, my concern about the complex number is the style of the visualization. It means when you print it out, we should have the real and imaginary values. Please double-check that part. The rest is good, thank you. Now, again, refresh to see... Okay, 25 is good. If some of you are not reaching the level of using the code that I think I shared on your Teams, make sure you're going to use this code. I'll look... I shared the code here for MSCA 100. You are accessing your Teams. Any questions? No? Let's go together for the campus tour. If you don't have questions, then we're going to continue tomorrow, possibly on class definition, defining more than one class in your project, and how we can change the structure of recording. We're going to wrap up our session a little bit sooner today, earlier than usual, because we're going to go together for a campus tour. All right. Let's leave your laptops here and then come with me and go down. Thank you.

2. STRUCTURAL REORGANIZATION (ENGLISH)

Mohammad: Did you install it? Because you are using Anaconda, and this is a pre-installed package.

`math` is a pre-installed package; you don't need to install it. If you need to install some target packages that are not available, we'll see that later on.

He also defined `maximum` as another function. Whether you use `maximum` as a function or not... `max(a, b)` is predefined; you just need to use it.

Okay, type conversion. You give it permission: if it's an integer, it will convert it to a float. If it's a float, leave it as it is. That's why we have this type conversion here, for two inputs, A and B.

Okay, we're going to have these as a complex, `z1` and `z2`, and we're going to return...

If you check this part, the result was correct for the complex. How do you print out the outcome of `add_complex`?

Is there any main function? Is there any main function in your project?

Yes, the compiler... If you are writing just a script, it's okay. But if you are writing a full project, companies look for the `main` function. We should add `main`; that is one part you can improve.

Also, my concern about the complex number is the style of the visualization. It means when you print it out, we should have the real and imaginary values. Please double-check that part. The rest is good, thank you.

Now, again, refresh to see... Okay, 25 is good.

If some of you are not reaching the level of using the code that I think I shared on your Teams, make sure you're going to use this code. I'll look... I shared the code here for MSCA 100. You are accessing your Teams.

Any questions?

No? Let's go together for the campus tour. If you don't have questions, then we're going to continue tomorrow, possibly on:

- Class definition
- Defining more than one class in your project
- How we can change the structure of recording

We're going to wrap up our session a little bit sooner today, earlier than usual, because we're going to go together for a campus tour.

All right. Let's leave your laptops here and then come with me and go down.

Thank you.

3. LOGICAL HIGHLIGHTING (ENGLISH)

Mohammad: Did you install it? Because you are using Anaconda, and this is a **pre-installed package**.

`math` is a **pre-installed package**; you don't need to install it. If you need to install some target packages that are not available, we'll see that later on.

He also defined `maximum` as another function. Whether you use `maximum` as a function or not... `max(a, b)` is **predefined**; you just need to use it.

Okay, **type conversion**. You give it permission: if it's an **integer**, it will convert it to a **float**. If it's a **float**, leave it as it is. That's why we have this **type conversion** here, for two inputs, A and B.

Okay, we're going to have these as a **complex**, `z1` and `z2`, and we're going to return...

If you check this part, the result was correct for the **complex** number. How do you print out the outcome of `add_complex` ?

Is there any **main function**? Is there any **main function** in your project?

Yes, the compiler... If you are writing just a **script**, it's okay. But if you are writing a **full project**, companies look for the **main function**. We should add `main`; that is one part you can improve.

Also, my concern about the **complex number** is the *style of the visualization*. It means when you print it out, we should have the **real** and **imaginary** values. Please double-check that part. The rest is good, thank you.

Now, again, refresh to see... Okay, 25 is good.

If some of you are not reaching the level of using the code that I think I shared on your Teams, make sure you're going to use this code. I'll look... I shared the code here for **MSCA 100**. You are accessing your **Teams**.

Any questions?

No? Let's go together for the **campus tour**. If you don't have questions, then we're going to continue tomorrow, possibly on:

- **Class definition**
- Defining more than one **class** in your project
- How we can change the structure of recording

We're going to wrap up our session a little bit sooner today, earlier than usual, because we're going to go together for a **campus tour**.

All right. Let's leave your laptops here and then come with me and go down.

Thank you.

4. INTERACTION EXTRACTION (ENGLISH)

(Note: This segment contains only the lecturer's monologue and instructions. There are no explicit student questions or responses captured in the provided transcript. The lecturer asks "Any questions?" but no student response is recorded.)

Mohammad: Did you install it? Because you are using Anaconda, and this is a pre-installed package. `math` is a pre-installed package; you don't need to install it. If you need to install some target packages that are not available, we'll see that later on. He also defined `maximum` as another function. Whether you use `maximum` as a function or not... `max(a, b)` is predefined; you just need to use it. Okay, type conversion. You give it permission: if it's an integer, it will convert it to a float. If it's a float, leave it as it is. That's why we have this type conversion here, for two inputs, A and B. Okay, we're going to have these as a complex, `z1` and `z2`, and we're going to return... If you check this part, the result was correct for the complex. How do you print out the outcome of `add_complex`? Is there any main function? Is there any main function in your project? Yes, the compiler... If you are writing just a script, it's okay. But if you are writing a full project, companies look for the `main` function. We should add `main`; that is one part you can improve. Also, my concern about the complex number is the style of the visualization. It means when you print it out, we should have the real and

imaginary values. Please double-check that part. The rest is good, thank you. Now, again, refresh to see... Okay, 25 is good. If some of you are not reaching the level of using the code that I think I shared on your Teams, make sure you're going to use this code. I'll look... I shared the code here for MSCA 100. You are accessing your Teams.

 **Mohammad:** Any questions? *(Pause)*  **Mohammad:** No? Let's go together for the campus tour. If you don't have questions, then we're going to continue tomorrow, possibly on class definition, defining more than one class in your project, and how we can change the structure of recording. We're going to wrap up our session a little bit sooner today, earlier than usual, because we're going to go together for a campus tour. All right. Let's leave your laptops here and then come with me and go down. Thank you.

5. SUMMARY & RESOURCES (ENGLISH)

Summary

Administrative Notices:

- Code for the course (**MSCA 100**) has been shared on the **Teams** platform. Students are expected to access and use it.
- The session concluded early to allow for a **campus tour**.
- The next session will continue with topics likely including **class definition**, defining multiple classes, and modifying the structure of recordings.

Core Academic Points:

- **Pre-installed Packages:** Emphasized that packages like `math` come pre-installed with distributions like Anaconda.
- **Type Conversion:** Discussed implicit type conversion in functions (e.g., integer to float).
- **Project Structure:** Highlighted the importance of including a `main` function in a full software project, as opposed to just a script, as it is a common industry expectation.
- **Code Quality & Output Formatting:** Stressed the need for proper visualization/output formatting, using the example of printing **complex numbers** with clear real and imaginary parts.

Recommended Learning Resources

- **Textbook:** *"Clean Code: A Handbook of Agile Software Craftsmanship"* by Robert C. Martin. Essential for understanding professional coding standards, project structure, and the importance of elements like the `main` function.

- **Official Documentation:** The Python documentation on [Modules](#) and [Python Packaging User Guide](#). Crucial for understanding pre-installed vs. installable packages and project organization.
- **Academic Paper/Concept:** *"Structuring Software for Reusability and Change"* (Parnas, 1979) or modern treatments on **Software Architecture**. Relevant for the upcoming discussion on defining multiple classes and structuring projects.
- **Style Guide: PEP 8 -- Style Guide for Python Code.** The definitive guide for Python code formatting and style, directly related to the "visualization" and readability concerns mentioned.

6. FULL CHINESE TRANSLATION

穆罕默德：你安装了吗？因为你使用的是 Anaconda，这是一个预安装的包。

`math` 是一个预安装的包；你不需要安装它。如果你需要安装一些不可用的目标包，我们稍后会看到。

他还定义了 `maximum` 作为另一个函数。无论你是否将 `maximum` 用作函数... `max(a, b)` 是预定义的；你只需要使用它。

好的，**类型转换**。你赋予它权限：如果它是**整数**，它将转换为**浮点数**。如果它是**浮点数**，则保持原样。这就是为什么我们在这里有**类型转换**，针对两个输入 A 和 B。

好的，我们将把这些作为**复数**，`z1` 和 `z2`，然后我们将返回...

如果你检查这部分，复数的结果是正确的。你如何打印出 `add_complex` 的结果？

有 `main` 函数吗？你的项目中有 `main` 函数吗？

是的，编译器... 如果你只是写一个**脚本**，那没关系。但如果你正在写一个**完整的项目**，公司会寻找 `main` 函数。我们应该添加 `main`；这是你可以改进的一个部分。

另外，我对**复数**的担忧是**可视化的风格**。这意味着当你打印它时，我们应该有**实部**和**虚部**的值。请仔细检查那部分。其余的都很好，谢谢。

现在，再次刷新看看... 好的，25 很好。

如果你们中有些人没有达到使用我认为我在你们 Teams 上分享的代码的水平，请确保你们会使用这个代码。我会看看... 我为 **MSCA 100** 在这里分享了代码。你们正在访问你们的 **Teams**。

🤖 穆罕默德：有什么问题吗？(停顿) 🤖 穆罕默德：没有？那我们一起去校园参观吧。如果你们没有问题，那么我们明天继续，可能涉及类的定义、在项目中定义多个类，以及我们如何改变录制的结构。我们今天要提前一点结束会议，比平时早，因为我们要一起去校园参观。

好了。把你们的笔记本电脑留在这里，然后跟我一起下楼。

谢谢。

总结

事务性通知汇总：

- 课程 (MSCA 100) 的代码已分享在 Teams 平台上。学生应访问并使用它。
- 会议提前结束以便进行校园参观。
- 下一次会议将继续讨论可能包括类的定义、定义多个类以及修改录制结构等主题。

核心学术知识点提炼：

- 预安装包：强调了像 `math` 这样的包随 Anaconda 等发行版预安装。
- 类型转换：讨论了函数中的隐式类型转换（例如，整数到浮点数）。
- 项目结构：强调了在完整的软件项目中包含 `main` 函数的重要性，而不仅仅是脚本，因为这是常见的行业期望。
- 代码质量与输出格式：强调了正确的可视化/输出格式的必要性，并以清晰打印复数的实部和虚部为例。

推荐学习资源

- 教科书：Robert C. Martin 所著的《代码整洁之道：敏捷软件工艺手册》。对于理解专业的编码标准、项目结构以及 `main` 函数等重要元素至关重要。
- 官方文档：Python 关于模块和 [Python 打包用户指南](#) 的文档。对于理解预安装包与可安装包以及项目组织至关重要。
- 学术论文/概念：“为可重用性和变更构建软件”(Parnas, 1979) 或关于软件架构的现代论述。与即将讨论的定义多个类和构建项目相关。
- 风格指南：PEP 8 -- Python 代码风格指南。关于 Python 代码格式和风格的权威指南，直接关系到所提到的“可视化”和可读性问题。